Price Prediction for Flights & 5* Hotels from NY Airports to Cancun

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1. **Project Description:** A machine learning predictive model to help maximize one's budget for a vacation in Cancun, utilizing trends to inform users on the best booking time for the best price and accommodations.

1.1. Problem Statement

The main issue that the creation of this project hopes to solve is the prediction of price fluctuations in real-time for booking flights to Cancun alongside a 5* hotel resort. Cancun is a world-famous travel destination, with countless people booking reservations months or years ahead of time, and the issue becomes timing the booking correctly based on a wide variety of factors, including but not limited to: plane prices, weather conditions, tourist density, crime rate, hotel promotions, and hotel/excursion accessibility. We aim to create a model that aggregates information together from the various factors that affect booking prices to produce both a compelling product and project, as well as build a useful tool to incorporate into future planning when traveling with friends and family.

1.2. Connection to Course Material

This project covers multiple topics from our course, including Python data structures, Numpy, Pandas, data frames, data collection and transformation, visualization, and predictive modelling. We used pandas to import the Cancun weather dataset, clean the data, rename columns, remove missing values, and organize the data into a dataframe. We used Numpy for most of the numerical work, for example, adding randomness to our synthetic flight and hotel prices to make them look more realistic. For predictive modelling, we use a Random Forest Regressor to predict prices based on factors such as weather, season, flight, and hotel prices.

2. **Novelty and Importance:**

2.1. Significance of the Project

International travel bookings are planned months in advance. With that in mind, how can travelers on a budget know the best time to book and what the weather conditions are during those months without extensive research? As a group of travelers who have spent hours creating one dubiously cost-effective itinerary after another, only for the price to drop further at some other time before or after we booked, the idea of a predictive model to better inform our decision-making was the perfect candidate. Combining personal experience and market trends with machine learning, a modeling system that indicates the timing of trips while accounting for conditions that can make or break an experience will offset the hours of research that go into traditional travel planning. Savvy users can derive insights that can save them hundreds of dollars and double their money's worth in relief from the stresses of travel planning.

2.2. Relevance

Traditional airline websites and hotel booking platforms mostly list current prices but fail to offer insights for when prices are likely to increase or decrease. They aren't incentivized to, as they would rather consumers buy travel packages when they are more expensive, for the sake of their own profit. Consequently, travelers often have to search manually and guess the best time to book their flights. By endeavoring to create a prediction model in this field, we are mechanizing the highly speculative world of travel pricing for a tool meant to serve consumers. Moreover, the hotel prices and flight prices can fluctuate due to seemingly unpredictable external factors such as hotel promotions, world events, and weather conditions. These fluctuations can be hard to detect, affect travel prices, and introduce noise to the prediction model while also affecting prediction accuracy.

3. Progress and Contribution:

3.1. Data Utilization

3.2. Models, Techniques, Algorithms

3.3. Experimental Design

Photo of Hotel CSV File:

	date	hotel_name	hotel_key	ota	price_usd
1	2026-04-18	GR Solaris Cancun	g150807-d501418	Booking.com	179
2	2026-04-18	GR Solaris Cancun	g150807-d501418	Trip.com	220
3	2026-04-18	GR Solaris Cancun	g150807-d501418	Agoda.com	171
4	2026-04-18	GR Solaris Cancun	g150807-d501418	Vio.com	166
5	2026-04-18	nds Cancun Resort & Spa	g150807-d671540		
6	2026-04-18	Live Aqua Cancun	g150807-d506374	Official Site	848
7	2026-04-18	All Inclusive - Adults Only	g150807-d303140		
8	2026-04-18	usive Resort - Adults Only	g150807-d15580232	Autograph	596
9	2026-04-18	usive Resort - Adults Only	g150807-d15580232	Trip.com	877
10	2026-04-18	usive Resort - Adults Only	g150807-d15580232	Agoda.com	579
11	2026-04-18	a Americana Coral Beach	g150807-d152681	Official Hotel	489
12	2026-04-18	a Americana Coral Beach	g150807-d152681	Booking.com	526
13	2026-04-18	a Americana Coral Beach	g150807-d152681	Trip.com	637
14	2026-04-18	a Americana Coral Beach	g150807-d152681	Agoda.com	526
15	2026-04-18	a Americana Coral Beach	g150807-d152681	Preferred Hotels	536
16	2026-04-18	lace The Grand - Cancún	g150807-d13138532	Booking.com	666
17	2026-04-18	lace The Grand - Cancún	g150807-d13138532	Trip.com	686
18	2026-04-18	lace The Grand - Cancún	g150807-d13138532	Agoda.com	696
19	2026-04-18	wn Paradise Club Cancun	g150807-d155814	Booking.com	244
20	2026-04-18	wn Paradise Club Cancun	g150807-d155814	Trip.com	285
21	2026-04-18	wn Paradise Club Cancun	g150807-d155814	Agoda.com	233
22	2026-04-18	wn Paradise Club Cancun	g150807-d155814	Vio.com	223
23	2026-04-18	emptation Cancun Resort	g150807-d154428	Booking.com	556
24	2026-04-18	emptation Cancun Resort	g150807-d154428	Official Site	549
25	2026-04-18	emptation Cancun Resort	g150807-d154428	Agoda.com	530
26	2026-04-18	Hotel Krystal Cancun	g150807-d154896	Booking.com	169
27	2026-04-18	Hotel Krystal Cancun	g150807-d154896	Official Site	154
28	2026-04-18	Hotel Krystal Cancun	g150807-d154896	Agoda.com	126
29	2026-04-18	Hotel Krystal Cancun	g150807-d154896	Vio.com	122

3.4. Key Findings and Results

3.5. Evaluation

3.6. Advantages and Limitations

4. Changes After Proposal:

4.1. Differences from Proposal

The main difference that derived from the Proposal was the belief in idealistic websites for data gathering and analysis. Initially, the collection of data seemed possible from open-source websites to determine current prices of hotels, flights, and weather distribution. However, after looking into the provided information of the sites themselves, many kept them behind locked paywalls. This caused a shift in focus in how data would be accumulated. For weather data, past historical API data had to be utilized. For hotels, a third-party API data-gathering source was needed, where a member would have to manually enter all the keys and create a CSV data file from scratch. Flight prices also relied on non-traditional sources, gathering data from outside sources that delved into actual flight patterns.

4.2. Challenges

One major difficulty we faced was collecting reliable flight prices. First, we tried using the Bureau of Transportation Statistics because it provides official transportation datasets, but the datasets were extremely large and not directly useful, as they mainly focused on domestic U.S. flights. Due to this, we used Google Flights API to extract future flight prices, giving us a more realistic idea of price trends. However, the API had usage limits, so we were only able to collect flight prices for June and July 2026, which were then used as a reference to create synthetic flight prices.

5. Conclusion and Future Work:

5.1. Summary of Contribution

Bismah was responsible for the collection of flight data and a great deal of the final report. This was an important role as the flight data is a difficult piece of data to collect due to limited access and large datasets. She experimented with multiple websites and data sources, but faced challenges to find relevant data so to overcome this she wrote API based code to gather real flight prices. Furthermore, due to her well-versed analytical and writing skills, she was able to fill a lot of the reports.

Gavin was responsible for the main code that gathered information from the flight, hotel, and weather reports to develop a machine learning algorithm that would predict prices considering all the factors together. With stronger coding ability, he decided to take on the responsibility alongside insight from other members to work on the main piece of the project.

Alvin was responsible for the discovery and application of weather historical data and hotel prices. With hotels also being locked behind private data, he had to search through and manually input all the hotel keys, developing a coding program to iterate through each to produce a manageable CSV file. Furthermore, with a program that would take multiple hours, he took on the role to run it behind the scenes for the development of a precise CSV file.

5.2. Future Direction

For future improvements, we can incorporate more real-world data instead of relying on synthetic data. Furthermore, if continued, the use of actual data sets can be purchased and utilized within the code to determine prices more accurately. Currently, a lot of machine learning and estimation is being derived as the price fluctuates wildly for hotel prices, weather, and flight prices. As such, if those averages are more concise within the data sets, the analysis by said machine is able to be fully optimized and imported into realistic production.